Supplementary material for "Thermodynamics of CO_2-loaded aqueous amines with ePC-SAFT. Part II: transfer to MDEA, AMP, piperazine and DEA"

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S1 Shared and amine-specific inputs

The common Helmholtz-energy contributions, combining rules, solvent permittivity expression and reference conversions are defined in the companion Part I [1]. The tables here reproduce scalar inputs from the actual parameter files used for these calculations. The complete electronic files also retain temperature-dependent coefficients, association-site multiplicities, correlations and the parameter ties. Shared component and pair values remain fixed; an interaction between a new amine species and a shared component can be fitted without changing the shared component itself. No fitted ionic size is interpreted as an independent molecular measurement.

Water has a temperature-dependent segment diameter,

$$\sigma_w(T) = 2.7804326 + 10.11\exp(-0.01775T) - 1.417\exp(-0.01146T) \quad \text{in Å, with } T \text{ in K.} \tag{S1}$$

Only the constant term in that law was fitted in Part I; the exponential coefficients were fixed. Water's segment number, dispersion and association parameters were independently fitted to IAPWS-95; CO_2 size and dispersion follow Gross and Sadowski [2, Table 2]. Inorganic-ion size and energy follow Uyan et al., with their hydrogen-ion description used for hydronium [3, Table 4]. The shared Born diameters and the water interactions other than the fitted bicarbonate term are assumptions specified in Part I, rather than inorganic-ion size measurements. The bicarbonate–water interaction is fixed at the MEA fit's upper bound. The MEA joint-fit R2 coefficients retain their conditional uncertainty in Part I, but no uncertainty of those shared inputs is propagated here.

Table S1
Shared species inputs. Constant scalar coefficients are shown; temperature laws, site topology, ties and fitted quantities are specified in the text and electronic parameter files.

Species	Quantity	Value	Unit
CO_2	ϵ/k_B	169.21	K
CO_2	m^{seg}	2.0729	1
CO_2	σ	2.7852	Å
CO_2	f	1	1
CO_2	d_V	1×10^{-8}	L/mol
CO_2	d_T	1403	K
water	ϵ/k_B	432.37881	K
water	m^{seg}	1.2527538	1
water	f	1.5	1
water	d_V	0.3777	L/mol
water	d_T	1403	K
HCO_3^-	d^{Born}	3	Å
HCO_3^-	d^{DH}	2.9296	Å
HCO_3^-	ϵ/k_B	70	K

Species	Quantity	Value	Unit
HCO_3^-	d^{pack}	2.578048	Å
HCO_3^-	m^{seg}	1	1
HCO_3^-	σ	2.9296	Å
HCO_3^-	f	1	1
CO_3^{2-}	d^{Born}	3	Å
CO_3^{2-}	d^{DH}	2.4422	Å
CO_3^{2-}	ϵ/k_B	249.26	K
CO_3^{2-}	d^{pack}	2.149136	Å
CO_3^{2-}	m^{seg}	1	1
CO_3^{2-}	σ	2.4422	Å
CO_3^{2-}	f	1	1
H_3O^+	d^{Born}	1.218	Å
H_3O^+	d^{DH}	3.4654	Å
H_3O^+	ϵ/k_B	500	K
H_3O^+	d^{pack}	3.049552	Å
H_3O^+	m^{seg}	1	1
H_3O^+	σ	3.4654	Å
H_3O^+	f	1	1
OH^-	d^{Born}	3.0810769	Å
OH^-	d^{DH}	2.0177	Å
OH^-	ϵ/k_B	650	K
OH^-	d^{pack}	1.775576	Å
OH^-	m^{seg}	1	1
OH^-	σ	2.0177	Å
OH^-	f	1	1

Table S2
Shared nonzero constant pair coefficients and explicit association edges. Unlisted constant pair corrections are zero. Temperature-law coefficients are stated separately.

Pair	Quantity	Value
CO_2–water	k ij	−0.0464169
CO_2–water	k ij reciprocal temperature slope	−35.6650522
HCO_3^-–water	k ij	1.0000000
CO_3^{2-}–water	k ij	−0.2500000
H_3O^+–water	k ij	0.2500000
OH^-–water	k ij	−0.2500000
water–water	association energy / K	1756.6497965
water–water	association volume	0.0376197
CO_2–water	association energy / K	878.3248982
CO_2–water	association volume	0.0376197
CO_2–water	association energy / K	878.3248982
CO_2–water	association volume	0.0376197

S1.1 Amine-specific species and association

MDEA retains its neutral pure-component and caloric description, its dielectric correlation and the fixed dissociation law. Its final fit releases the neutral–water interaction and linear temperature slope, cation interactions with water, CO_2 and bicarbonate, a reciprocal-temperature slope for cation–water, and cation segment diameter, dispersion energy and Born diameter. The segment and Debye–Hückel diameters are tied. The neutral–water law is $k(T) = k_0 + k_1(T - 313.15\,\mathrm{K})$, with $k_0 = -0.2063373482$ and $k_1 = 0.0014750401\ \mathrm{K}^{-1}$. The cation–water law is $k(T) = k_0 + b_k(1/T - 1/313.15\,\mathrm{K})$ with $k_0 =$

-0.03323620783 and $b_k = -27.10492507$ K. The nine coordinates do not include a fitted dissociation constant.

Table S3
MDEA amine-specific species inputs. Constant scalar coefficients are shown; temperature laws, site topology, ties and fitted quantities are specified in the text and electronic parameter files.

Species	Quantity	Value	Unit
MDEA	m^{seg}	3.675	1
MDEA	σ	3.563	Å
MDEA	ϵ/k_B	228.711	K
MDEA	f	1	1
MDEA	d_V	0.86306697	L/mol
MDEA	d_T	1121.7735	K
$MDEAH^+$	m^{seg}	3.675	1
$MDEAH^+$	σ	3.6494074	Å
$MDEAH^+$	ϵ/k_B	623.04303	K
$MDEAH^+$	f	1	1
$MDEAH^+$	d^{DH}	3.6494074	Å
$MDEAH^+$	d^{Born}	7	Å

Table S4
MDEA nonzero constant pair coefficients and explicit association edges. Unlisted constant pair corrections are zero. Temperature-law coefficients are stated separately.

Pair	Quantity	Value
CO_2–$MDEAH^+$	k ij	0.2481889
water–$MDEAH^+$	k ij	−0.0332362
water–$MDEAH^+$	k ij reciprocal temperature slope	−27.1049251
HCO_3^-–$MDEAH^+$	k ij	1.0000000
MDEA–MDEA	association energy / K	2046.6240000
MDEA–MDEA	association volume	0.1180069
$MDEAH^+$–$MDEAH^+$	association energy / K	0.0000000
$MDEAH^+$–$MDEAH^+$	association volume	0.1180069

AMP uses its neutral association description with explicit carbamate chemistry and a fixed Hamborg dissociation law. The two temperature-dependent AMP–water and $AMPH^+$–water interactions have the reciprocal-temperature form above, with slopes -55.96106619 and -274.1548086 K, respectively. Its final nine coordinates include those four interaction quantities, cation–CO_2 interaction, cation and carbamate diameters, hydrolysis level and hydrolysis enthalpy correction. Ion Debye–Hückel diameters equal segment diameters and packing diameters equal 0.88σ. The fitted hydrolysis level and enthalpy correction are completed together about 313.15 K, so the tabulated completed-law level differs from the fitting coordinate's level. The carbamate diameter is fixed at its lower bound for the conditional uncertainty calculation.

Table S5
AMP amine-specific species inputs. Constant scalar coefficients are shown; temperature laws, site topology, ties and fitted quantities are specified in the text and electronic parameter files.

Species	Quantity	Value	Unit
AMP	ϵ/k_B	237.05992	K
AMP	m^{seg}	4.0761318	1
AMP	σ	3.1870454	Å

Species	Quantity	Value	Unit
AMP	f	1	1
AMP	d_V	0.32290165	L/mol
AMP	d_T	2055.8807	K
$AMPH^+$	d^{Born}	5	Å
$AMPH^+$	d^{DH}	5.4152589	Å
$AMPH^+$	ϵ/k_B	237.05992	K
$AMPH^+$	d^{pack}	4.7654278	Å
$AMPH^+$	m^{seg}	1	1
$AMPH^+$	σ	5.4152589	Å
$AMPH^+$	f	1	1
carbamate	d^{Born}	5	Å
carbamate	d^{DH}	5.3199952	Å
carbamate	ϵ/k_B	207.21695	K
carbamate	d^{pack}	4.6815957	Å
carbamate	m^{seg}	1	1
carbamate	σ	5.3199952	Å
carbamate	f	1	1

Table S6
AMP nonzero constant pair coefficients and explicit association edges. Unlisted constant pair corrections are zero. Temperature-law coefficients are stated separately.

Pair	Quantity	Value
CO_2–$AMPH^+$	k ij	0.7549873
AMP–water	k ij	−0.1307372
AMP–water	k ij reciprocal temperature slope	−55.9610662
carbamate–water	k ij	−0.0036580
$AMPH^+$–water	k ij	−0.3136388
$AMPH^+$–water	k ij reciprocal temperature slope	−274.1548086
AMP–AMP	association energy / K	2385.1893365
AMP–AMP	association volume	0.0366211

PZ uses an effective two-site neutral association description; the pure fit holds segment count 2.6519 and diameter 3.7136 Å fixed while fitting its three energy/association quantities to Steele pressures. Pure-liquid density and the liquid extrapolation below the melting region remain unvalidated. The neutral PZ–water interaction has the linear-temperature form anchored at 353.15 K, with $k_0 = -0.1976454016$ and $k_1 = 0.0011250148\ \mathrm{K}^{-1}$; its two coordinates were fitted separately to binary pressure and vapor composition, then fixed during the reactive fit. The reactive twelve-coordinate fit ties the singly and doubly protonated cation sizes and energies; mono- and dicarbamate share size, energy and water interaction. Four charged PZ species have fixed 5 Å Born diameters. The neutral HPZCOO zwitterion has independent size and energy and no association sites. The remaining fitted quantities are the cation–water and cation–bicarbonate interactions and three carbamate-law offsets. PZ and HPZCOO dielectric inputs are transferred from MEA as an assumption.

Table S7
PZ amine-specific species inputs. Constant scalar coefficients are shown; temperature laws, site topology, ties and fitted quantities are specified in the text and electronic parameter files.

Species	Quantity	Value	Unit
PZ	m^{seg}	2.6519	1
PZ	σ	3.7136	Å

Species	Quantity	Value	Unit
PZ	ϵ/k_B	313.20598	K
PZ	f	1	1
PZ	d_V	0.73098659	L/mol
PZ	d_T	1019.0803	K
PZH^+	m^{seg}	1	1
PZH^+	σ	4.2856111	Å
PZH^+	ϵ/k_B	102.72806	K
PZH^+	d^{Born}	5	Å
PZH^+	d^{pack}	3.7713378	Å
PZH^+	d^{DH}	4.2856111	Å
PZH^+	f	1	1
PZH_2^{2+}	m^{seg}	1	1
PZH_2^{2+}	σ	4.2856111	Å
PZH_2^{2+}	ϵ/k_B	102.72806	K
PZH_2^{2+}	d^{Born}	5	Å
PZH_2^{2+}	d^{pack}	3.7713378	Å
PZH_2^{2+}	d^{DH}	4.2856111	Å
PZH_2^{2+}	f	1	1
PZ carbamate	m^{seg}	1	1
PZ carbamate	σ	4.86247	Å
PZ carbamate	ϵ/k_B	717.58778	K
PZ carbamate	d^{Born}	5	Å
PZ carbamate	d^{pack}	4.2789736	Å
PZ carbamate	d^{DH}	4.86247	Å
PZ carbamate	f	1	1
PZ dicarbamate	m^{seg}	1	1
PZ dicarbamate	σ	4.86247	Å
PZ dicarbamate	ϵ/k_B	717.58778	K
PZ dicarbamate	d^{Born}	5	Å
PZ dicarbamate	d^{pack}	4.2789736	Å
PZ dicarbamate	d^{DH}	4.86247	Å
PZ dicarbamate	f	1	1
HPZCOO	m^{seg}	2.6519	1
HPZCOO	σ	3.8142737	Å
HPZCOO	ϵ/k_B	975.42757	K
HPZCOO	f	1	1
HPZCOO	d_V	0.73098659	L/mol
HPZCOO	d_T	1019.0803	K

Table S8
PZ nonzero constant pair coefficients and explicit association edges. Unlisted constant pair corrections are zero. Temperature-law coefficients are stated separately.

Pair	Quantity	Value
water–PZH^+	k ij	−1.0000000
water–PZH_2^{2+}	k ij	−1.0000000
water–PZ carbamate	k ij	0.0740702
water–PZ dicarbamate	k ij	0.0740702
HCO_3^-–PZH^+	k ij	−0.0210355
HCO_3^-–PZH_2^{2+}	k ij	−0.0210355
PZ–PZ	association energy / K	1469.8200923
PZ–PZ	association volume	0.0272299

DEA neutral pure parameters use Nasrifar and Tafazzol's four-site scheme, with two donor and two acceptor sites [4, Table 2, Eq. 34]. Their association strength uses $\sigma^3\kappa$, consistent with the present implementation. The fitted neutral DEA–water interaction is temperature independent, $k = -0.0636$, and the dielectric coefficients were fitted separately to unloaded permittivity observations before reactive estimation. The reported reactive fit represents $DEAH^+$ as a charged chain carrying DEA's segment count, diameter, dispersion energy and four sites, with zero self-association energy. The carbamate retains a single-sphere representation. Only the two ion–water interactions and hydrolysis level and enthalpy correction are fitted in the final four-coordinate objective. The alternative single-sphere cation fit is retained only as a comparison.

Table S9
DEA amine-specific species inputs. Constant scalar coefficients are shown; temperature laws, site topology, ties and fitted quantities are specified in the text and electronic parameter files.

Species	Quantity	Value	Unit
DEA	ϵ/k_B	269	K
DEA	m^{seg}	5.2048	1
DEA	σ	2.9477	Å
DEA	f	1	1
DEA	d_V	0.75748303	L/mol
DEA	d_T	1155.7835	K
$DEAH^+$	d^{Born}	5	Å
$DEAH^+$	d^{DH}	2.9477	Å
$DEAH^+$	ϵ/k_B	269	K
$DEAH^+$	m^{seg}	5.2048	1
$DEAH^+$	σ	2.9477	Å
$DEAH^+$	f	1	1
carbamate	d^{Born}	5	Å
carbamate	d^{DH}	6.1672093	Å
carbamate	ϵ/k_B	183.00063	K
carbamate	d^{pack}	5.4271442	Å
carbamate	m^{seg}	1	1
carbamate	σ	6.1672093	Å
carbamate	f	1	1

Table S10
DEA nonzero constant pair coefficients and explicit association edges. Unlisted constant pair corrections are zero. Temperature-law coefficients are stated separately.

Pair	Quantity	Value
DEA–water	k ij	−0.0636335
carbamate–water	k ij	0.0057148
$DEAH^+$–water	k ij	−0.2151637
DEA–DEA	association energy / K	1517.6000000
DEA–DEA	association volume	0.1014000
$DEAH^+$–$DEAH^+$	association energy / K	0.0000000
$DEAH^+$–$DEAH^+$	association volume	0.1014000

S1.2 Reaction directions, coefficients and reference states

The shared reactions are water autoprotolysis, $CO_2 + 2\,H_2O \rightleftharpoons HCO_3^- + H_3O^+$, and $HCO_3^- + H_2O \rightleftharpoons CO_3^{2-} + H_3O^+$. Amine dissociation is written $AH^+ + H_2O \rightleftharpoons A + H_3O^+$; carbamate hydrolysis is written $ACOO^- + H_2O \rightleftharpoons A + HCO_3^-$. PZ adds successive protonation, monocarbamate

and dicarbamate reactions and zwitterion proton transfer, as specified in the electronic stoichiometric file. The R1 and R3 source constants have a mole-fraction infinite-dilution basis, while R2 and the amine laws have a common-molality basis. The tables retain those source coefficients rather than silently substituting a different basis. The thermodynamic standard-state conversion uses water as pure-liquid solvent and neutral solutes at infinite dilution, following Part I. For a fitted enthalpy correction H about T_p, the native completed-law correction is $\delta a = H/(RT_p)$ and $\delta b = -H/R$, preserving $\ln K(T_p)$. This correction is part of the reaction law and is not an independently added absorption-heat term.

Table S11
MDEA reaction coefficients in $\ln K = a + b/T + c\ln(T/313.15\,\mathrm{K}) + dT$. Coefficients a, c are dimensionless, b is K and d is K^{-1}. Source-reference basis is retained, not silently interchanged. PZ P3–P5 include the fitted offsets.

Reaction	a	b	c	d	Basis
R1	3.729098	−13445.9	−22.4773	0	mole fraction
R2	23.87092	−12017.43	−36.7816	0	molality
R3	12.14579	−12431.7	−35.4819	0	mole fraction
MDEA dissociation	−16.39891	−819.7	10.976	0	molality

Table S12
AMP reaction coefficients in $\ln K = a + b/T + c\ln(T/313.15\,\mathrm{K}) + dT$. Coefficients a, c are dimensionless, b is K and d is K^{-1}. Source-reference basis is retained, not silently interchanged. PZ P3–P5 include the fitted offsets.

Reaction	a	b	c	d	Basis
R1	3.729098	−13445.9	−22.4773	0	mole fraction
R2	23.87092	−12017.43	−36.7816	0	molality
R3	12.14579	−12431.7	−35.4819	0	mole fraction
AMP dissociation	−8.973099	−3850.736	8.064629	0	molality
carbamate hydrolysis	8.210491	−1944.838	0	0	molality

Table S13
PZ reaction coefficients in $\ln K = a + b/T + c\ln(T/313.15\,\mathrm{K}) + dT$. Coefficients a, c are dimensionless, b is K and d is K^{-1}. Source-reference basis is retained, not silently interchanged. PZ P3–P5 include the fitted offsets.

Reaction	a	b	c	d	Basis
R1	3.729098	−13445.9	−22.4773	0	mole fraction
R2	23.87092	−12017.43	−36.7816	0	molality
R3	12.14579	−12431.7	−35.4819	0	mole fraction
P1	−11.55919	−3123.533	6.786961	0	molality
P2	−2.804785	−2824.703	3.563322	0	molality
P3	8.791731	−3615.98	0	0	molality
P4	−10.04478	−3493.022	0	0	molality
P5	3.392648	−1322.144	0	0	molality

Table S14
DEA reaction coefficients in $\ln K = a + b/T + c\ln(T/313.15\,\mathrm{K}) + dT$. Coefficients a, c are dimensionless, b is K and d is K^{-1}. Source-reference basis is retained, not silently interchanged. PZ P3–P5 include the fitted offsets.

Reaction	a	b	c	d	Basis
R1	3.729098	−13445.9	−22.4773	0	mole fraction
R2	23.87092	−12017.43	−36.7816	0	molality
R3	12.14579	−12431.7	−35.4819	0	mole fraction

Reaction	a	b	c	d	Basis
DEA dissociation	−9.279878	−4214.076	0	0.009961213	molality
carbamate hydrolysis	5.926332	−2631.648	0	0	molality

MDEA's Kamps–Maurer measurements span 278–368 K (printed p. 1509); the higher computational declaration does not extend the measurements. PZ carbamate measurements are at 283.15–333.15 K, even though their rounded source-law interval is written 283–333 K. DEA's Bower law is measured only to approximately 323 K, while its computational declaration reaches 393.15 K. The AMP hydrolysis slope is fitted, while the independent fixed dissociation declaration refuses temperatures above 353.22 K. All coefficient tables must be read together with these source ranges and main-text Table 2.

S2 Observation inventory and fitting objectives

The observation roles in main-text Table 3 apply throughout this supplement. No pressure set labeled held out enters any objective, including earlier fits shown for sensitivity. Prior scoring is nonetheless material: MDEA outside-fit sets were already inspected, AMP Kundu was available before final diameter choice, PZ Ermatchkov before the penalty alternative and final fit choice, and DEA holdouts before a numerically tied second-start choice. Comparison-only DEA properties were added after selection. A datum's absence from a final objective is not evidence of complete blindness during model development.

Table S15
Measurement locators preserved with the calculation inputs. Source row identities and original-observable columns accompany the electronic numerical files; fitted and unfitted roles are in the main paper.

Observation	Source	Locator
MDEA Kuranov	Kuranov1996	Table 2, printed p. 1960
MDEA Jakobsen	Jakobsen2005	Table A5, PDF p.9, printed p.9902
MDEA Carson	Carson2000	Table 3, PDF p.9, printed p.1293
MDEA Merkley	Merkley1987	Figure 3; PDF p.5 printed p.441
MDEA Cp	Weiland1997	Table 1, PDF p.1, printed p.1004
MDEA loaded-density	Weiland1998	Table 1, PDF p.2, printed p.379
MDEA unloaded-density	Sobrino2016	Table 5, PDF p.4, printed p.234
MDEA Kim	Kim2008	Table 8, PDF p.4, printed p.2524
AMP loaded-pressure	Hartono	Source rows and definitions in electronic data
AMP binary	Hartono	Source rows and definitions in electronic data
AMP nmr	Ciftja 2014	Source rows and definitions in electronic data
DEA SeoHong	SeoHong1996	Table 1 (DEA/AMP = 30/0), PDF p.2, printed p.259
DEA Lal	Lal1985	Table 2, PDF p.2, printed p.682
DEA weiland	Weiland1998	Table 2, PDF p.2, printed p.379
DEA lawson viii	LawsonGarst1976	Table VIII (CO2 only; 77 F rows 54.5 wt% per footnote a), PDF p.10, printed p.29
DEA bottinger DEACOO	Bottinger2008	Table 4, PDF p.8, printed p.138
DEA cai	Cai1996	Table 4, PDF p.2, printed p.1102
DEA excess enthalpy	Horstmann2002	Table 6, PDF p.3, printed p.1498

For MDEA, Kuranov total pressures use natural-log ratios, scale one and weight 100/9; Kim binary total pressures use natural-log ratios with unit scale and weight. Jakobsen protonated fractions use differences scaled by 10 % of the observed fraction, with unit weight. Carson and Merkley heat differences use 3000 J/mol CO_2 scales and unit weights. Weiland heat capacities use 2 % of the observed heat capacity and unit weights. Loaded-density differences use the stored row-specific scales and weight 1/33, whereas unloaded densities use 0.5 kg/m^3 and weight ten. Thus the dimensionless Kuranov scaled RMS is not its physical pressure AARD, and these group costs are not common physical accuracy measures. The

stored rows preserve loading caps for integral heat, source amount bases for heat capacity and density, and unavailability of capped states.

AMP loaded total pressures use log ratios with unit scale and weight $100/[9(47)\max(u_P/P, 0.05)^2]$. The fifty-two binary entries comprise twenty-six pressures and twenty-six vapor AMP fractions; binary log pressures use 0.05 scale through their weight, while vapor fractions use $\max(10^{-4}, 0.05y)$. The binary group is averaged over its fifty-two entries. Ciftja carbamate fractions are mapped to carbamate per total amine, with difference scale 0.003 and group weight $1/57$. The pressure, binary and species groups are not pooled when reporting their physical deviations. Kundu's 18, 25 and 30 mass-percent stocks correspond nominally to 2.0, 2.8 and 3.4 mol/L and span loading 0.412–0.966. Kundu's loading uncertainty is not a combined partial-pressure uncertainty, so it cannot supply an uncertainty-normalized pressure test.

PZ pressure differences use an operational scale 0.02 times observed pressure. NMR normalization propagates independent area errors with $q = 0.02$ into correlated group-fraction blocks; complete-state areas reconstruct the monocarbamate pair and free/dicarbamate groups before normalization. Blank dicarbamate observations are not zeros and do not create additional residuals. The resulting 232 observations come from 121 source states and use a full block covariance. The assumed area scale is not a source-validated independent species standard deviation. Source group fractions are per total PZ, while source loading per twice the PZ amount is converted to the per-PZ convention before the bubble calculation.

DEA minimizes

$$C = \frac{1}{2}\left[\frac{1}{30}\sum_{j=1}^{30}\left(\frac{\ln(p_j^c/p_j^o)}{0.09}\right)^2 + \frac{1}{50}\sum_{j=1}^{50}\left(\frac{\Delta h_j^c - \Delta h_j^o}{0.02|\Delta h_j^o|}\right)^2\right]. \tag{S2}$$

Lawson pressures cover 310.93–380.37 K and loading no lower than 0.376 at a single fitted concentration. Carson heats cover mass fractions 0.10, 0.20 and 0.30 at 298.15 K. Integral absorption enthalpy is $(H_f - H_i - \Delta n_{CO_2} h_{CO_2}^{feed})/\Delta n_{CO_2}$, with equilibrium liquids and feed evaluated at the same temperature and pressure; negative values are exothermic. It is not a differential heat and is not the excess enthalpy per mole of binary mixture.

S3 Starts, selected parameters and calibration residuals

Table S16
Original search bounds of the reported reactive fits. Reaction-level bounds refer to the fitted coordinate before the coupled enthalpy completion; DEA enthalpy bounds are refit increments. PZ values and ties follow the complete twelve-coordinate chart. Bounds are not measured molecular intervals.

Amine	Coordinate	Lower	Upper
MDEA	pair/mdea-water/k_ij/a0: temperature law coefficient	−0.5	0.5
MDEA	pair/mdea-water/k_ij/a1: temperature law coefficient	−0.01	0.01
MDEA	$MDEAH^+$–CO_2: k ij	−0.5	0.5
MDEA	$MDEAH^+$–water: k ij	−0.5	0.5
MDEA	$MDEAH^+$–water: k ij reciprocal temperature slope	−1000	1000
MDEA	$MDEAH^+$–HCO_3^-: k ij	−1	1
MDEA	$MDEAH^+$: segment diameter	3	4.5
MDEA	$MDEAH^+$: dispersion energy over k	50	950
MDEA	$MDEAH^+$: born diameter	2	7
AMP	AMP–water: k ij	−0.5	0.5
AMP	AMP–water: k ij reciprocal temperature slope	−1000	1000
AMP	$AMPH^+$–water: k ij	−0.5	0.5
AMP	$AMPH^+$–water: k ij reciprocal temperature slope	−1000	1000
AMP	$AMPH^+$–CO_2: k ij	−1	1
AMP	$AMPH^+$: segment diameter	4.5	6.5

Amine	Coordinate	Lower	Upper
AMP	carbamate: segment diameter	5.32	6.5
AMP	carbamate hydrolysis a	8.555	13.555
AMP	carbamate hydrolysis enthalpy increment / J mol^{-1}	−5000	5000
DEA	$DEAH^+$–water: k ij	−1	1
DEA	carbamate–water: k ij	−1	1
DEA	carbamate hydrolysis a	4.56444	10.5644
DEA	carbamate hydrolysis enthalpy increment / J mol^{-1}	−19000	21000
PZ	Cation diameter / Å	2	9
PZ	Cation dispersion / K	50	1000
PZ	Cation–water interaction	−1	0.5
PZ	Cation–bicarbonate interaction	−1	0.5
PZ	Carbamate diameter / Å	2	9
PZ	Carbamate dispersion / K	50	1000
PZ	Carbamate–water interaction	−0.5	0.5
PZ	HPZCOO diameter / Å	2	6
PZ	HPZCOO dispersion / K	50	1000
PZ	P3, P4 and P5 offsets (each)	−3	3

Table S17
Retained start outcomes. Costs use different objective scales and must not be compared across amines. The MDEA A3 rows and smaller-diameter AMP fits are sensitivity comparisons, not additional selected results.

Fit / start	Cost	Iterations	Bounds	Termination
MDEA A3 endpoint	403.57153	10	4	converged
MDEA A3 alternative	403.57153	11	4	converged
MDEA B1 endpoint	374.51908	11	2	converged
MDEA B1 alternative	392.92418	17	2	converged
AMP reported floor	61.73030	45	2	converged
AMP 3.5 Å start 1	61.61537	54	2	converged
AMP 3.5 Å start 2	61.61537	52	2	converged
AMP 3.5 Å start 243	61.61537	28	2	converged
PZ reported	20698.620	100	1	iteration limit
DEA A-1	10.139543	8	0	converged
DEA A-2	10.139543	8	0	converged
DEA B-1	7.970700	6	0	converged
DEA B-2	7.970700	7	0	converged

MDEA B1 retains the nine-coordinate fit under fixed shared inputs. Its selected cost is 374.5190788 for 278 objective rows, with two active bounds; the second declared start converges to a larger cost, so agreement of two starts is not claimed. The A3 alternative is the same-data sensitivity counter-case, with no replacement of the reported B1 fit. B1's Carson scaled RMS increases from 0.97662 to 1.10594, a documented caloric compromise rather than a hidden failure relabeled as improved agreement. Its selected parameter-file SHA-256 is `2a5341be176829d087133a1467828f762a655f92a586eb1062a5719a58142d9d`.

AMP uses the scored fit with floor 5.32 Å, cost 61.73030, rather than the lower-cost 3.5 Å alternative. All three starts of the latter reach the new smaller-diameter bound; the small cost change on restoring the larger floor shows weak identification below about 5.3 Å. The reported floor is the MEA carbamate diameter, while segment-volume scaling suggests a larger AMP carbamate, approximately 5.89 Å. Neither estimate is a measured ionic diameter. The completed hydrolysis enthalpy correction reaches −5000 J/mol in every refit of the sequence.

PZ's reported twelve-coordinate fit starts from the earlier fit projected onto the new ties. The independent proposed start is unavailable with HPZCOO included in the vapor; no successful fit from that start is implied. The reported fit reaches 100 iterations with cost 20698.6 nearly flat over its final fourteen

iterations. A conditional diagnostic with the bound coordinate fixed converges at a cost lower by approximately 3×10^{-4}, but does not retroactively label the original termination converged. PZH^{+}–water reaches the replacement lower bound -1. The separate penalty alternative is not substituted for this reported fit.

DEA compares two cation representations with two starts each. The two single-sphere starts give cost 10.13954, and the two charged-chain starts give 7.97070, all without active bounds. The charged-chain starts differ in cost by 4.8×10^{-10}, below the function tolerance, and their largest relative prediction difference is 2.5×10^{-6}. The selected second start therefore represents the same numerical optimum as the first to the stated resolution. Its held-out sets were first scored on the first start and then on this tied second start, whose endpoint was frozen before rescoring. An unavailable more negative cation–water start is not counted as a converged solution.

Main-text Fig. 1 displays the fitted-pressure residual shapes. The electronic row files retain all fitted pressure, heat and species predictions, including zeros and the actual observed-value denominators. PZ NMR error is reported in group fraction and whitened residual units; DEA NMR below is in true mole fraction. The same percentage formula does not make these different species observables comparable.

S4 Unfitted and comparison-only properties by source and range

Main-text Tables 6 and 5 retain source, pressure type, temperature blocks and error signs. The Ermatchkov MDEA total is a mixed-range summary, while every Xiao row is within the 278–368 K dissociation measurements. Kuranov's thirty extrapolated fitted rows are at 373.14–373.17 K and 393.12–393.15 K. The fitted Merkley heats give 10.36 % AARD at 333.15 K and 6.56 % at 388.71 K, each over two observations; signed errors are -5725 and -3989 J/mol CO_2, respectively. The latter is a dissociation-law extrapolation. Stored Carson, heat-capacity and loaded-density rows do not carry temperature fields sufficient for a law-range aggregate, so none is fabricated. Binary pressure and unloaded density do not depend on the reactive dissociation calculation.

AMP Kundu temperature blocks are 303, 313 and 323 K, with twenty-five, fifteen and eleven observations. Their pressure biases are +53.94, +43.78 and -7.09 %, respectively, whereas Tong's seventeen total-pressure observations span approximately 313, 333 and 353 K. Tong's twenty-three higher-temperature loaded rows are outside the fixed dissociation declaration and have no finite score here. The source-law temperature basis of hydrolysis is extrapolated for the Kundu cold block even though the fitted completed hydrolysis law is computationally declared there. These distinct law limits preclude one blanket in-range label for AMP.

PZ Ermatchkov has nineteen partial pressures at 313.1 K, eighteen at 353.15 K and fifteen at 393.15 K. Only the first block lies within the measured carbamate interval; the middle block already extrapolates carbamate. At the hottest temperature all fifteen calculated pressures are high. Eight observations at the lowest temperature, three at the middle and none at the highest lie within the printed uncertainty. The fifty-two stored pointwise source uncertainties are distinct from parameter uncertainty.

DEA Seo and Hong has five, six and five pressures at 40, 60 and 80 °C; Lal has seventeen and twenty-seven at 40 and 100 °C. Only the 40 °C blocks are within the Bower measured range. Lal states an accuracy of ±15 % above 0.05 kPa and much larger uncertainty below that pressure; five of twenty-eight higher-pressure rows fall within that stated interval. The sixteen lower-pressure observations remain in the overall AARD. At 100 °C, four very-low-loading values are described by Lal as probably in error (printed p. 684); their exclusion is not used to improve the reported overall score.

DEA comparison-only Lawson Table VIII contains ten observations: two at 77 °F, four at 100 °F and four at 200 °F. The nominally fifty-mass-percent solution is 54.5 wt% in its coldest block. Cai's nine binary observations compare the model bubble pressure at measured liquid composition and temperature with 6.66 kPa. Weiland's forty-three loaded densities at 298 K exclude the documented misprinted twenty-mass-percent, loading-0.50 value. These comparisons did not enter the reactive objective or final parameter selection.

The Böttinger DEA comparison separates nine states without the oxazolidone byproduct from sixteen with byproduct. For the clean states, amine sum, carbamate and bicarbonate-sum mean absolute

errors are 0.001749, 0.001301 and 0.001274 in true mole fraction. For byproduct-containing states the corresponding values are 0.003166, 0.004986 and 0.005347, and omitted byproduct chemistry prevents treating these as equivalent tests. The model uses assumed 2.5 MPa rather than unavailable individual cell pressures. No uncertainty for the NMR mole fractions is supplied, so these absolute errors are not uncertainty-normalized validation scores.

The fourteen Horstmann excess enthalpies are at 363.15 K and 1.75 MPa, with source scale $s = 2\ \mathrm{J/mol} + 0.01|H^E|$. The model gives positive mixing heat at every state while all observations are negative. The ideal-gas enthalpy contribution is linear in composition and cancels in excess enthalpy, which is calculated from the liquid residual enthalpies. The stored pure-water residual enthalpy is about -41.4 kJ/mol as a unit check. This is a comparison-only binary caloric result, separate from Carson's fitted absorption heats. No analogous assessment for the other amines establishes whether the error is specific to DEA.

S5 Finite phase searches and unavailable states

Main-text Table 8 reports the finite state inventories rather than a phase diagram. For MDEA, distinct observation identities can represent repeated physical states; 775 records should not be described as 775 independent experiments. The thirty-seven gas witnesses are not dense-liquid splits, and all seven corrected-Kamps domain exclusions remain outside the phase evidence. Kuranov source rows 28 and 72 exceed shared R2's 393.15 K upper limit and are excluded from the 278-row fit. Neither historical computational templates ending at 395 K nor older-law calculations extend the current R2 interval. The capped MDEA heat at 333.15 K, 60 wt% and 1.121 MPa remains unavailable on the original search, with no restricted loading grid used to manufacture a value.

AMP nominal Kundu states are searched at their calculated bubble states; loading-perturbed pressure evaluations do not supply additional phase evidence. Its calibration gas outcomes were not stored and cannot be reconstructed as zero from the absence of a reported value. DEA's 140 searched fitted/held-out states comprise thirty pressures, fifty heats, sixteen Seo and Hong and forty-four Lal states. The additional comparison-only sets are not included in that count. No additional liquid or gas phase was detected in those DEA searches, and no global-stability conclusion follows.

PZ searches 121 NMR liquid states and sixty-two Hilliard bubble states. Where an assumed 0.1 MPa NMR liquid gives a gas-like split, it is moved to 1.05 times its calculated bubble pressure before the liquid check. The reported fit has negative tangent-plane distances at Hilliard rows 55, 56, 61 and 62 of approximately -0.096, -0.111, -0.0024 and $-0.031\ RT$. These are false liquid splits at the observed liquid compositions, not experimental phase-separation measurements. Solid equilibrium is absent, including at temperatures below the pure-PZ melting region.

The separate penalty fit uses the Part I tangent-plane requirement at 245 states: the 121 NMR liquids and each Hilliard feed as an all-liquid parent at 1.05 and three times its candidate bubble pressure. Penalty scales are 10^{-2} followed by 3×10^{-3}. Its routine search reports one split, but direct distance calculations at split compositions recovered near the fit show two further negative values at rows 55 and 56. Those direct values, $-0.0309\ RT$ and $-0.0437\ RT$, demonstrate that the finite search missed resolved splits. The penalty did not establish a single-liquid fit. Consequently detected-split counts are lower bounds throughout both papers' interpretation here, and a count of states without detected instability is an upper bound on stability within the searched inventory, not proof of global stability.

S6 Conditional covariance, sensitivity and bound profiles

All parameter errors are local and conditional on the species model, fixed shared inputs, data weights and active bounds. They exclude experimental systematic errors, future measurement noise, model discrepancy and uncertainty in transferred constants. Covariance propagation includes parameter correlations, while the maximum one-SE sensitivity in the tables moves one parameter marginally with the others

fixed. For a bound parameter, the last column instead uses the first inward profile step with other parameters refitted. Those perturbation sizes are different; their common display is not a statistical significance ranking. Parameter units are those of Section S1, and enthalpy shifts are J/mol.

Table S18
MDEA conditional parameter errors and largest fractional prediction change. The last column uses one marginal standard error for interior parameters and the first inward refit step for bound parameters; these perturbations have different sizes. Units follow the named parameters; enthalpy shifts are J/mol.

Parameter	Value	SE	Largest change
$MDEAH^+$: ϵ/k_B	623.04	36.686	0.87587
$MDEAH^+$, water: k_ij	−0.033236	0.0099899	0.66959
$MDEAH^+$: σ	3.6494	0.030615	0.096832
$MDEAH^+$, water: water interaction slope	−27.105	3.3628	0.090821
MDEA–water a0: coefficient	−0.20634	0.0013412	0.044204
MDEA–water a1: coefficient	0.001475	2.5631×10^{-5}	0.041912
$MDEAH^+$, CO_2: k_ij	0.24819	0.029345	0.0046476
$MDEAH^+$, HCO_3^-: k_ij	1	–	0.0042142
$MDEAH^+$: born_diameter	7	–	0.0013609

Table S19
AMP conditional parameter errors and largest fractional prediction change. The last column uses one marginal standard error for interior parameters and the first inward refit step for bound parameters; these perturbations have different sizes. Units follow the named parameters; enthalpy shifts are J/mol.

Parameter	Value	SE	Largest change
$AMPH^+$: σ	5.4153	0.43273	0.62404
$AMPH^+$, water: k_ij	−0.31364	0.038238	0.52636
$AMPH^+$, water: water interaction slope	−274.15	85.929	0.12653
AMP, water: k_ij	−0.13074	0.0060466	0.11814
AMP, water: water interaction slope	−55.961	15.867	0.033161
$AMPH^+$, CO_2: k_ij	0.75499	0.087274	0.030051
carbamate-hydrolysis/a	8.2105	0.29086	0.020545
carbamate: σ	5.32	–	0.00039001
carbamate-hydrolysis/total enthalpy shift	−5000	–	3.5166×10^{-5}

Table S20
PZ conditional parameter errors and largest fractional prediction change. The last column uses one marginal standard error for interior parameters and the first inward refit step for bound parameters; these perturbations have different sizes. Units follow the named parameters; enthalpy shifts are J/mol.

Parameter	Value	SE	Largest change
PZH^+/u	102.73	21.234	1.6031
HPZCOO/u	975.43	113.63	1.0564
carbamate/u	717.59	141.82	0.89897
HPZCOO/sigma	3.8143	0.066159	0.80996
PZH^+/bicarbonate-k	−0.021035	1.031	0.78658
PZH^+/sigma	4.2856	0.32334	0.51166
carbamate/sigma	4.8625	0.35603	0.50409
P4/offset	−0.019326	0.13257	0.47204

Parameter	Value	SE	Largest change
carbamate/water-k	0.07407	0.049728	0.3681
P3/offset	0.15704	0.12251	0.35292
P5/offset	−0.26155	0.060658	0.058636
PZH^+/water-k	−1	–	0.0031208

Table S21
DEA conditional parameter errors and largest fractional prediction change. The last column uses one marginal standard error for interior parameters and the first inward refit step for bound parameters; these perturbations have different sizes. Units follow the named parameters; enthalpy shifts are J/mol.

Parameter	Value	SE	Largest change
carbamate-hydrolysis/a	5.9263	0.5973	0.57223
carbamate, water: k_ij	0.0057148	0.052851	0.46078
$DEAH^+$, water: k_ij	−0.21516	0.0034456	0.083032
carbamate-hydrolysis/total enthalpy shift	−12216	1309.3	0.071383

MDEA has two active bounds and conditional variance factor 2.7639784. The inward cation–bicarbonate interaction profiles give cost increases 0.142, 0.7282 and 1.508 at values 0.98, 0.90 and 0.80. Born-diameter profiles at 6.95, 6.75 and 6.50 Å give increases 0.06927, 0.3571 and 0.7433. All converge, but this limited grid does not locate a confidence limit or a global minimum. The two weakest full-Jacobian directions include both bound parameters, with singular values 0.00848001 and 0.0198683 in the original optimizer units. Their inclusion of constrained coordinates limits interpretation of two-sided displacements.

AMP has two bound coordinates and variance factor 0.829. The $AMPH^+$ diameter and cation–water interaction have correlation about 0.96; cation–water interaction and slope have correlation about 0.98. Inward carbamate diameter steps of 1, 5 and 10 % of the original interval give cost changes 0.0009, 0.0044 and 0.0087. The corresponding enthalpy-correction steps give changes 0.0063, 0.032 and 0.064. The twelve retained AMP profile fits, including the interior cation–CO_2 interaction diagnostic, converge. Nearly flat bound profiles restrict molecular interpretation even when covariance bands are comparatively small.

PZ holds one bound coordinate fixed and uses eleven interior coordinates with variance factor 146.3. The conditional refit moves enough that the endpoint inverse-Jacobian calculation is used rather than identifying its covariance with the original fit without qualification. The 1 % inward profile converges; 5 and 10 % profiles stop at their forty-iteration budget, with cost changes 46.5 and 61.9. Those two changes are upper bounds on minimized profile cost. The weakest full-chart singular direction has singular value about 0.21 per optimizer unit and is dominated by cation–bicarbonate and cation–water interactions. Large marginal pressure sensitivities make its linear estimates indicative rather than reliable finite-displacement predictions.

DEA has no active bounds and uses variance factor 0.210. Its hydrolysis-law level and carbamate–water interaction have correlation 0.972, so an individual marginal sensitivity can exceed the correlated prediction error. The source-separated pressure SEs are much smaller than the held-out errors, and fitted heat SEs are only 0.9–1.9 %. This is conditional information within the chosen model; it does not apportion the remaining error between experiments and thermodynamic formulation.

S7 AMP substitution sequence and source-comparison qualifications

The complete AMP sequence uses the same fifty-one Kundu observations at every step, with no Kundu residual entering any objective. The earlier shared inputs give 43.69 % Kundu AARD and +40.51 %

bias. Substituting Part I's shared bicarbonate–water interaction and R2 without refitting gives 21.89 % and −15.10 %, while worsening the fitted total pressures. Refitting reverses that apparent improvement, as shown in Table S22. This is a numerical attribution under the specified substitutions, not an identified physical defect or proof that either experimental source is wrong.

Table S22
AMP same-data sequence. Calibration uses total pressure and carbamate fraction; Kundu uses partial pressure and remains outside every objective. The fourth row is the reported fit.

Change	Cost	Fitted P AARD / %	Species RMS	Kundu AARD / %	Bias / %
Shared substitution; no AMP refit	238.38	18.85	0.01386	21.89	−15.10
Original AMP limits	66.28	9.93	0.00614	43.51	40.35
Carbamate diameter floor raised	66.42	9.93	0.00637	43.03	39.77
Cation–CO2 interaction widened	61.73	10.33	0.00631	42.85	37.79
Carbamate diameter floor lowered	61.62	10.34	0.00586	43.95	39.09

The earlier bicarbonate–water interaction is 0.72705584775, compared with one in Part I. The earlier R2 coefficients $a = 23.58551614210$ and $b = -11928.05673746$ K change to $a = 23.87092013090$ and $b = -12017.43099656$ K, while $c = -36.7816$ remains fixed in the same molality basis. The six common component descriptions do not otherwise change in this sequence. The comparisons are same-data parameter substitutions, not an independent physical-validation experiment.

A separate data-only comparison of Hartono and Kundu lacks sufficient overlap to decide whether the systematic Kundu error reflects experiment or the model. There is no adequate 303 K overlap, and the exploratory interpolation requires concentration and vapor-pressure-baseline assumptions. Its exploratory ratios are therefore not used as independent quantitative evidence in this paper. The open source-versus-model question remains alongside the measured forty-three-percent error, rather than being replaced by a favorable earlier score.

S8 Numerical identities and display provenance

The four studies use the merged ePC-SAFT source commit `391dfe5d64ede5036bf160bc6ae037ad5e14af32` and immutable wheel SHA-256 `0962e21d2fa7a8ba1ed4622f1110b3e8053bacd878fa41c999d85e8b9afab66b`. The common MEA parameter input has SHA-256 `f8f64820b161377a0d97d1e4ad4f6cea764cd0558dc63231d58e2096988eef97`. The electronic input index binds the exact amine-specific files and laws to those shared inputs. Stored replay checks verify numerical reproduction and balance; they are not physical validation.

MDEA replay reproduces all 278 fitted predictions with zero stored prediction difference and relative cost difference 2.76×10^{-13}. AMP's scored fit replays cost within 7.4×10^{-13} relative. DEA cold reload reproduces cost within 8×10^{-13} relative and weighted residuals within 1.3×10^{-11}. PZ retains its iteration-limited fit, conditional diagnostic, residuals and all phase trials, including the direct split witnesses missed in the penalty fit's search. Numerical precision does not account for the larger experimental discrepancies.

The presentation script reads stored JSON and CSV values without importing the thermodynamic model. Every generated table records input SHA-256 values in its source comments. The accompanying numerical-source index maps table cells and manuscript paragraph values to source files and fields; figure values and parameter-only bands remain separate from experimental uncertainties. All plots use discrete observations and model evaluations at the measured states; no interpolated model sweep or new phase boundary is introduced. The bibliography snapshot is shared with Part I, with a separate entry for the unpublished companion manuscript.

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